Can one hear the shape of a random matrix?

Elia Bisi

(University of Florence)

UNIVERSITÀ DEGLI STUDI FIRENZE

Discrete Random Structures II

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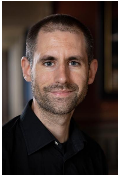
Papers and collaborators

- E. Bisi and F. D. Cunden, **λ-shaped random matrices, λ-plane trees, and λ-Dyck paths**. Electron. J. Probab. 30 (2025).
- E. Bisi, F. D. Cunden, I. Hartarsky, and S. Wagner, **Can one hear the shape of a random matrix?**. In preparation (2026+).

Joint work with:

Fabio D. Cunden
(Bari)

Ivailo Hartarsky
(Lyon)



Stephan Wagner
(Graz)

Overview

1. Classical random matrices
2. λ-shaped random matrices
3. Limiting spectral measure
4. Hearing the shape
5. Conclusions

Classical random matrices

- $\{X_{ij} : i, j \geq 1\}$ – complex i.i.d., with $\mathbb{E}X_{ij} = 0$ and $\mathbb{E}|X_{ij}|^2 = 1$
- $X_N = (X_{ij})_{i,j=1}^N$ – $N \times N$ random matrix
- $W_N = \frac{1}{N} X_N X_N^*$ – $N \times N$ Hermitian random matrix
- $x_1 \geq x_2 \geq \cdots \geq x_N \geq 0$ – (random) eigenvalues of W_N
- $\varrho_N = \frac{1}{N} \sum_{i=1}^N \delta_{x_i}$ – empirical spectral measure

Problem

Does W_N have a **limiting spectral measure** as $N \to \infty$?

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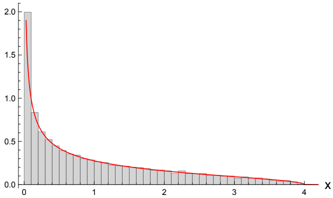
Marchenko–Pastur distribution and Catalan numbers

Theorem (Marchenko–Pastur, 1967)

As $N \to \infty$, almost surely, the spectral measure ϱ_N converges weakly to

$$\varrho_{\mathrm{MP}}(\mathrm{d}x) = \frac{1}{2\pi}\sqrt{\frac{4-x}{x}}\,\mathbb{1}_{[0,4]}(x)\,\mathrm{d}x.$$



- ϱ_{MP} is the (unique) prob. meas. on $[0,\infty)$ whose moments are the Catalan numbers:

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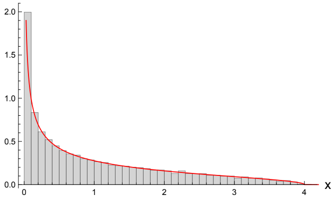

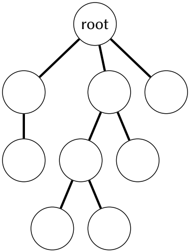

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Integer partitions ≡ Young diagrams

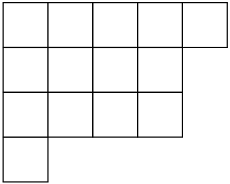
Young diagram: $\lambda = (5, 4, 4, 1)$

Length: $\ell(\lambda) = 4$

Size: $|\lambda| = 14$

Diagonal sequence: $D^{\lambda} = (1, 2, 3, 4, 3, 1)$

Staircase size: $q(\lambda) = 4.$



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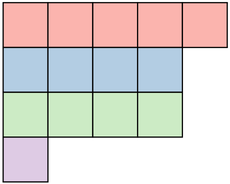

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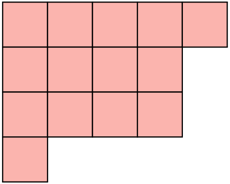

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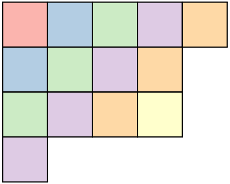

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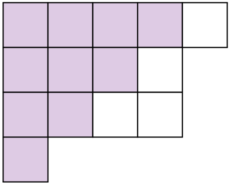

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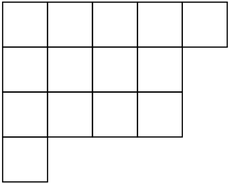

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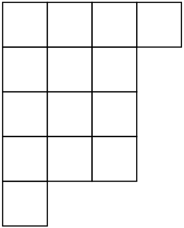
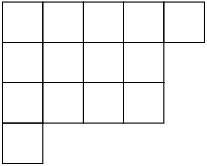


λ-shaped random matrices

- Young diagram λ:



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$$X = \begin{pmatrix} X_{11} & X_{12} & X_{13} & X_{14} & X_{15} \\ X_{21} & X_{22} & X_{23} & X_{24} & 0 \\ X_{31} & X_{32} & X_{33} & X_{34} & 0 \\ X_{41} & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad X_{ij}\text{'s i.i.d.}$$

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What are the spectral properties of the $\ell \times \ell$ matrix XX^* in the limit of large shapes (here $\ell := \ell(\lambda)$)?

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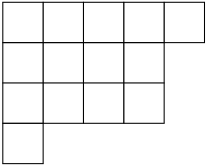

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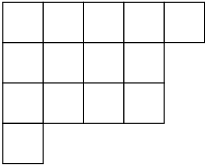

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Rank of λ-RMs

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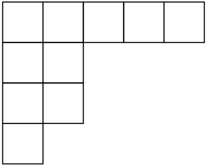
$\lambda =$ 

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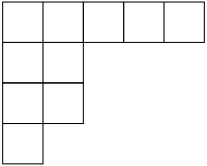
$\lambda =$ 

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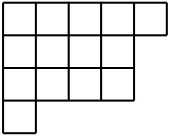
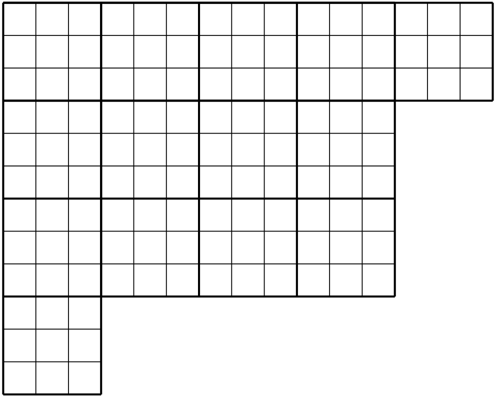
$3 \cdot$  $=$ 

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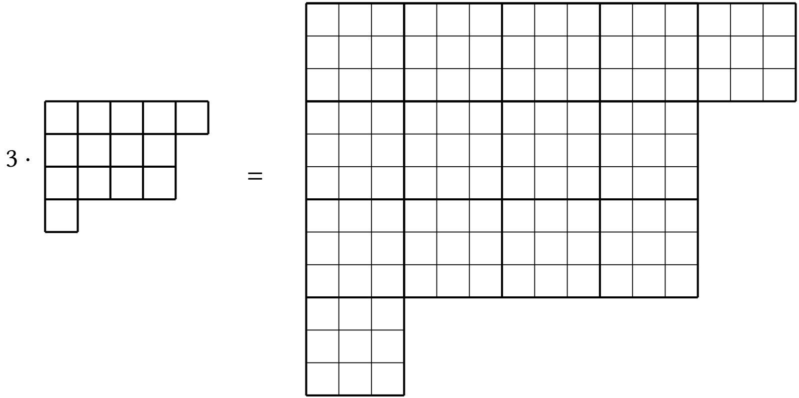

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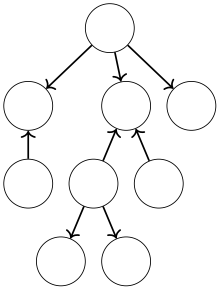
$$\mathbb{E}\int x\,\mathrm{d}\varrho_N^\lambda(x) = \mathbb{E}\frac{1}{Nq}\sum_{i=1}^{Nq} x_i = \frac{1}{Nq}\mathbb{E}\mathrm{Tr}W_N = \frac{1}{N^2q^2}\sum_{(i,j)\in N\lambda}\mathbb{E}|X_{ij}|^2 = \frac{N^2|\lambda|}{N^2q^2}$$

λ-plane trees

A λ-**plane tree** is a pair (T, c), where:

- T is a rooted plane tree with vertex set $V = V_0 \sqcup V_1$ (even and odd vertices) and directed edge set $E \subseteq V_0 \times V_1$
- $c : V \to \mathbb{N}$ is a labelling such that $(c(v_0), c(v_1)) \in \lambda$ for all $(v_0, v_1) \in E$



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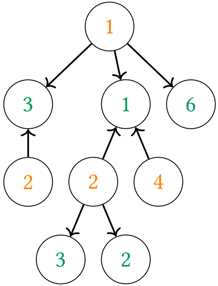

$\lambda =$

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	
(4,1)					

Limiting spectral measure

Theorem (B.–Cunden–Hartarsky–Wagner, 2026+)

As $N \to \infty$, almost surely, ϱ_N^{λ} converges weakly to the probability measure ϱ^{λ} whose moments are

$$\int_{\mathbb{R}_{\geq 0}} x^k \varrho^{\lambda}(x)dx = \frac{C_k^{\lambda}}{q^{k+1}},$$

where

$$C_k^{\lambda} := \#\{\lambda\text{-plane trees on } k+1 \text{ vertices}\}.$$

- When $\lambda = (1) = \square$, we have $C_k^{\lambda} = C_k$, and ϱ^{λ} is the Marchenko–Pastur distribution
- For $k = 1$, we have $\frac{C_1^{\lambda}}{q^2} = \frac{|\lambda|}{q^2}$

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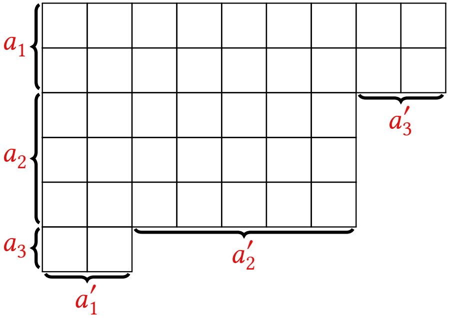
Multiplicities of parts in a partition

- $a_1, \ldots, a_r$: multiplicities of the parts of λ
- $a'_1, \ldots, a'_r$: multiplicities of the parts of λ'



Enumeration of λ-plane trees

Theorem (B.–Cunden–Hartarsky–Wagner, 2026+)

Let λ has multiplicities $\boldsymbol{a} = (a_1, \dots, a_r)$ and λ' has multiplicities $\boldsymbol{a}' = (a'_1, \dots, a'_r)$, then

$$C_k^{\lambda} = \sum_{\substack{\boldsymbol{e},\boldsymbol{o} \in \mathbb{Z}_{\geq 0}^r : \\ |\boldsymbol{e}|+|\boldsymbol{o}|=k+1}} s(e_1, \dots, e_r, o_1, \dots, o_r) \prod_{i=1}^{r} a_i^{e_i} a_i'^{o_i},$$

$$s(e_1, \dots, e_r, o_1, \dots, o_r) = \frac{1}{k} \prod_{j=1}^{\lceil r/2 \rceil} \binom{e_{\geq j} + o_{\leq r-j+1} - 1}{e_j} \binom{o_{\geq j} + e_{\leq r-j+1} - 1}{o_j}$$
$$\times \prod_{j=1}^{\lfloor r/2 \rfloor} \binom{o_{\leq j} + e_{\geq r-j+1} - 1}{e_{r-j+1}} \binom{e_{\leq j} + o_{\geq r-j+1} - 1}{o_{r-j+1}}.$$

Proof: based on generating functions and Lagrange inversion formula

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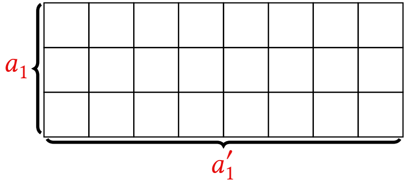
Rectangular matrices

Special case $r = 1$:
$a_1 \times a_1'$ matrices

$\lambda =$



$$C_k^{\lambda} = \sum_{m=1}^{k} N(k,m) {a_1}^{m} {a_1'}^{k+1-m}$$

$$N(k,m) = \frac{1}{k}\binom{k}{m}\binom{k}{m-1} = s(m, k+1-m) \qquad \text{(Narayana numbers)}$$

- $s(e_1, \ldots, e_r, o_1, \ldots, o_r)$ are multivariate generalisations of Narayana numbers
- limiting spectral distribution: Marchenko–Pastur law with parameter a_1/a_1'

Can one hear the shape of a random matrix?

- X_N: $(N\lambda)$-shaped random matrix
- Y_N: $(N\mu)$-shaped random matrix
- Suppose they have the same limit law $\varrho^\lambda = \varrho^\mu$
- Is it true that $\lambda = \mu$?

Question

Is the map $\lambda \mapsto \varrho^\lambda$ injective?
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Answer

No. There exist distinct partitions λ, μ s.t. $\varrho^{\lambda} = \varrho^{\mu}$.

Examples

Examples where $\varrho^{\lambda} = \varrho^{\mu}$:

- *dilations*: $\lambda = \ydiagram{2}$, $\mu = \ydiagram{4,4}$
- *conjugations*: $\lambda = \ydiagram{3,1}$, $\mu = \ydiagram{2,1,1}$
- minimal non-trivial example: $\lambda = \ydiagram{3,1}$, $\mu = \ydiagram{2,2}$
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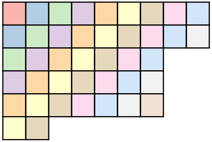
Diagonal profile

$\lambda =$ 

Diagonal sequence of λ:

$$D^{\lambda} = (1, 2, 3, 4, 5, 6, 6, 5, 5, 3, 1)$$

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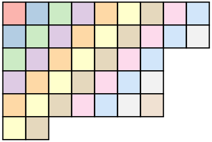
$\lambda =$ 

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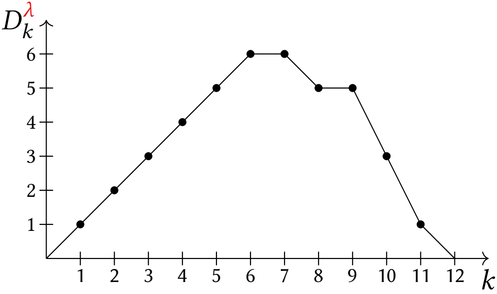

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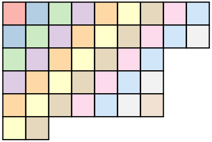
$\lambda =$ 

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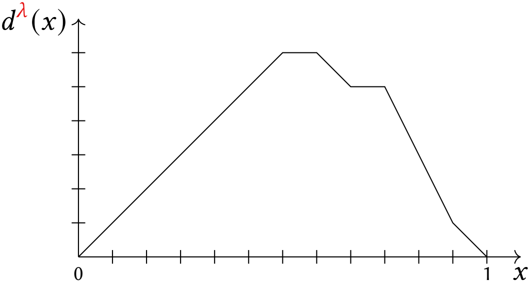
Diagonal profile of λ:

$$d^{\lambda} : [0, 1] \to [0, 1]$$



Limiting spectral measures and diagonal profiles

Theorem (B.–Cunden–Hartarsky–Wagner, 2026+)

Let λ and μ be two integer partitions. Then,

$$\varrho^{\lambda} = \varrho^{\mu} \quad \Longleftrightarrow \quad d^{\lambda} = d^{\mu}.$$

- The following Young diagrams are 'isospectral' with law ϱ_{MP}:

- The following Young diagrams are not 'isospectral':

$\lambda =$, $\mu =$

$$D^{\lambda} = (1,2,3,4,2,1,1) \qquad D^{\mu} = (1,2,3,4,3,1)$$

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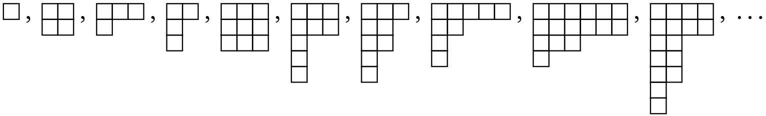

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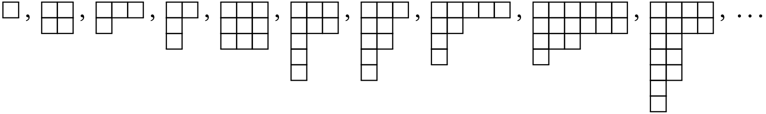

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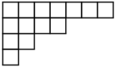
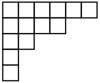
$\lambda =$  , $\mu =$ 

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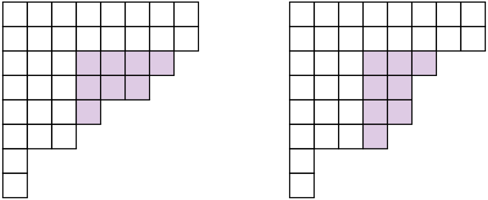
Key ideas

Partial conjugation:



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Two Young diagrams λ, μ have the same diagonal sequence if and only if there exists a sequence of partial conjugations taking λ to μ.

Theorem

Let λ and μ be two Young diagrams that differ by a single partial conjugation. Then, $C_k^\lambda = C_k^\mu$ for all $k \geq 1$.

Proof: explicit bijection between λ-plane trees and μ-plane trees.

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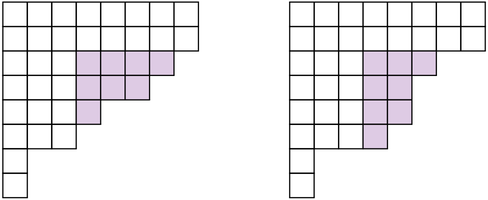

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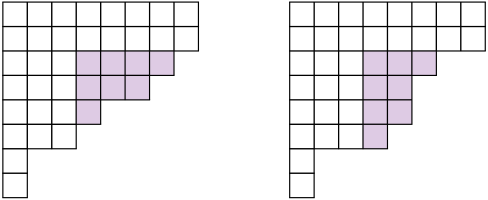

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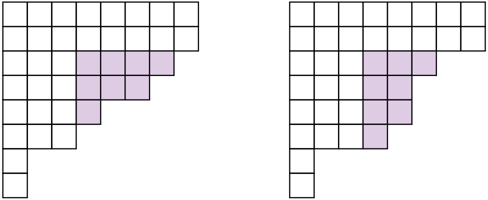

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Conclusions

Take-home messages:

- $(N\lambda)$-shaped random matrices have a limiting spectral distribution whose moments enumerate λ-**plane trees**.
- Explicit formulas for the enumeration problem.
- No, you cannot hear the shape of a random matrix... but **you can hear the diagonal profile** of the shape!

Work in progress:

- General sequence of shapes converging to a limit shape.
- Integrable λ-shaped random matrices.

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